

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B. Tech. (CE) Examination****April / May 2018****CE210 MICROPROCESSOR & COMPUTER ARCHITECTURE****Date: 09.05.2018, Wednesday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 Answer the following questions. [06]**
- a. Why 8085 is called as 8-bit Processor? [01]
 - b. How many Data Lines & Address Lines available in 8085? [02]
 - c. Give the Example of 1-Byte, 2-Byte & 3-Byte Instruction of 8085 [03]
- Q – 2 Answer the following questions. (Any Three) [15]**
- a. Explain Opcode Fetch Machine Cycle [05]
 - b. Explain Addressing Modes of 8085 [05]
 - c. Write a short note on various Data transfer instructions. [05]
 - d. Explain Various Flags of 8085 [05]
 - e. Write an Assembly Language Program to Add two 8-bit numbers stored in memory locations 4001H and 4006H, and store the result at 400AH location. [05]
- Q - 3 Answer the following questions. (Any two) [14]**
- a. Draw & Explain Functional block diagram of 8085 [07]
 - b. Draw the Timing Diagram of MVI A, 32H instruction. PC = 2000 H, Opcode = 3E H [07]
 - c. Give the Difference between Memory Mapped I/O & I/O Mapped I/O [07]
 - d. Interface 4Kb EPROM with 8085. Also give the Starting & Ending Address of EPROM [07]

SECTION – II

- Q - 4 Answer the following questions. [06]**
- a. Which Interrupt is having Highest Priority? [01]
 - b. What is the difference between JUMP & CALL instruction in 8085? [02]
 - c. What is the Difference between RLC & RAL instruction of 8085? Explain it by suitable diagram [03]
- Q –5 Answer the following questions. (Any Three) [15]**
- a. Explain the working of PUSH & POP instruction with diagram [05]
 - b. Explain the working of DMA controller [05]
 - c. Explain the Interrupts of 8085 [05]
 - d. Write an Assembly Language Program to Multiply two 8-bit numbers stored in memory locations 4001H and 4006H, and store the result at 400AH location. [05]

- e. Write an Assembly Language Program to check whether the given number at location 5000 H is even or odd. If the number is even, then store it at 2000 H else store it at 2001H [05]

Q – 6 Answer the following questions. (Any Two) [14]

- a. Explain Nesting of Subroutine with suitable example [07]
- b. Draw & Explain block diagram of 8155 [07]
- c. Draw & Explain block diagram of 8255 [07]
- d. Write an Assembly Language Program to control Traffic Light for the given time period. Green Light ON time = 15 Seconds, Yellow Light ON time = 5 Seconds, Red Light ON time = 20 Seconds [07]